Lec-3

11/10/26 10:02 AM

Ques 21:
n = 5

```
*                               *
*   *                       *   *
*   *   *               *   *   *
*   *   *   *       *   *   *   *
*   *   *   *   *   *   *   *   *
```

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        int row = 1;
        int star = 1;
        int space = 2 * n - 3;
        while (row <= n) {
            // star
            int i=1;
            while(i<=star) {
                System.out.print("* ");
                i++;
            }
            // space
            int j=1;
            while(j<=space) {
                System.out.print("  ");
                j++;
            }
            // star
            int k=1;
            while(k<=star) {
                System.out.print("* ");
                k++;
            }
            // next ki prep
            row++;
            System.out.println();
            star++;
            space -= 2;
        }
    }
```

N = 5, Row = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5

star = ~~1~~ ~~2~~ ~~3~~ ~~4~~ 5 space = ~~7~~ ~~5~~ ~~3~~ ~~1~~ -1

Ques 22:
n = 5

```
*   *   *   *   *   *   *   *   *
*   *   *   *       *   *   *   *
*   *   *               *   *   *
*   *                       *   *
*                               *
```

1 3 5 5

Row=1 star=N space=-1

while (Row<=n){
//star
//space
//star
//next line
space+2
star--

}

Ques 14:
n = 5

```
                *
            *   *
        *   *   *
    *   *   *   *
*   *   *   *   *
    *   *   *   *
        *   *   *
            *   *
                *
```

→M
→N

Row=1 space=N-1 star=1

while(Row<=2xN-1){
//space
//star
//mirror
//next line
sysout()
Row++
}

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        int row = 1;
        int star = 1;
        int space = n - 1;
        while (row <= 2 * n - 1) {
            // space
            int i = 1;
            while (i <= space) {
                System.out.print("  ");
                i++;
            }
            // star
            int j = 1;
            while (j <= star) {
                System.out.print("* ");
                j++;
            }
            // mirror
            star++;
            space--;
            // next line prep
            row++;
            System.out.println();

        }
    }
```

Ques 14:
n = 5

```
                *
            *   *
        *   *   *
    *   *   *   *
*   *   *   *   *
    *   *   *   *
        *   *   *
            *   *
                *
```

N = 5, Row=1, star=~~1~~ ~~2~~ ~~3~~ ~~4~~ 5 4 3 2

space=~~4~~ ~~3~~ ~~2~~ ~~1~~ 0 ~~3~~ ~~2~~ ~~1~~ 4

Row=1, star++
Row=2 space--
Row=3
Row=4

Row=5 | star--
Row=6 | space++
Row=7
Row=8

Ques 16:
n = 5

```
        *   *   *   *   *
      *   *   *   *
    *   *   *
  *   *
*
  *   *
    *   *   *
      *   *   *   *
        *   *   *   *   *
```

Row=1 space=N-1 star=n

while(Row<=2xn-1) {
//space

//star

if(Row<n){
space--
star--
}
else
space++
star++
}
Row++
}

N=7

```
*   *   *       *   *   *
*   *               *   *
*                       *

*                       *
*   *               *   *
*   *   *       *   *   *
```

Row=1 star=N/2 space=1

while(Row<=n){
//star
//space
//star

if(Row<n/2+1){
Star--
space+2
}
else{
Star++
space-=2
}
sysout()
Row++
}

N=7 Row=1 star=N/2+1 space=-1

Ques 19:
n = 7

```
*   *   *   *   *   *   *
*   *   *       *   *   *
*   *               *   *
*                       *
*   *               *   *
*   *   *       *   *   *
*   *   *   *   *   *   *
```

→Mirror
N/2+1

while (Row<=n){
//star
//space
//star
mirror
if(Row<n/2+1){
star--
space+=2
}
else
star++
space-=2
}
Row++
s/y
}

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 7;
        int row = 1;
        int star = n / 2+1;
        int space = -1;
        while (row <= n) {
            // star
            int i = 1;
            while (i <= star) {
                System.out.print("* ");
                i++;
            }
            // space
            int j = 1;
            while (j <= space) {
                System.out.print("  ");
                j++;
            }
            // star
            int k = 1;
            while (k <= star) {
                System.out.print("* ");
                k++;
            }
            if (row < n / 2 + 1) {
                star--;
                space += 2;
            } else {
                star++;
                space -= 2;
            }
            row++;
            System.out.println();
        }
    }
```

1<4 4<4

Row=~~1~~ 7 6 5 4 3 2 star=~~4~~ 3 2 ~~1~~ 2 ~~3~~ 4 space=~~-1~~ ~~1~~ 3 5 3 1 -1

Row==1
k=2

Row==n
k=2

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        int row = 1;
        int star = 1;
        int space = n - 1;
        int val=1;
        while (row <= n) {
            // space
            int i = 1;
            while (i <= space) {
                System.out.print("  ");
                i++;
            }
            // star
            int j = 1;
            while (j <= star) {
                System.out.print(val+" ");
                val++;
                j++;
            }
            // Next Row ki prep
            star += 2;
            space--;
            row++;
            System.out.println();
        }
    }
```

val=~~1~~ ~~2~~ 5 10

Row=~~1~~ 2 3 star=~~1~~ 3 5 space=4 3 2 1 0

Ques 25:
n = 5

```
            1
         2  3  4
      5  6  7  8  9
  10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
```

Ques 27:
n = 5

```
                1
            1   2   1
        1   2   3   2   1
    1   2   3   4   3   2   1
1   2   3   4   5   4   3   2   1
```

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        int row = 1;
        int star = 1;
        int space = n-1;
        while (row <= n) {
            // space
            int i=1;
            while(i<=space) {
                System.out.print("  ");
                i++;
            }
            // star
            int j=1;
            int val=1;
            while(j<=star) {
                System.out.print(val+" ");
                val++;
                j++;
            }
            // Next Row ki prep
            star+=2;
            space--;
            row++;
            System.out.println();
        }
    }
```

Row=1

Ques 27:
n = 5

mirror

```
                1
            1   2   1
        1   2   3   2   1
    1   2   3   4   3   2   1
1   2   3   4   5   4   3   2   1
```

3/2+1 = ②

⑧

star=~~1~~ 3 5

```
// star
int j=1;
int val=1;
while(j<=star) {
    System.out.print(val+" ");
    if(j<star/2+1)
    val++;
    else
    Val--;
    j++;
}
```

1<2

Star+=2

```
        1
    2   3   2
3   4   5   4   3
    2   3   2
        1
```

```
        *
    *   *   *
*   *   *   *   *
    *   *   *
        *
```

```
        1
    2   2   2
3   3   3   3   3
    2   2   2
        1
```

Row=1 star=1 space=N-1

while (Row<=2xN-1){
//space

//star

//mirror
if(Row<n){
space--
star+=2
}
else{
space++
star-=2
}
Sysout
Row++
}

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 3;
        int row = 1;
        int star = 1;
        int space = n - 1;
        int val=1;
        while (row <= 2 * n - 1) {
            // space
            int i = 1;
            while (i <= space) {
                System.out.print("\t");
                i++;
            }
            // star
            int j = 1;

            while (j <= star) {
                System.out.print(val+"\t");
                j++;
            }

            // Mirror
            if (row < n) {
                star += 2;
                space--;
                val++;
            } else {
                star -= 2;
                space++;
                val--;
            }
            // Next row ki prep
            row++;
            System.out.println();
        }

    }
```

2<3 3<3

Row=~~1~~ ~~2~~ ~~3~~ star=~~1~~ ~~3~~ ~~5~~ ~~3~~ 1 space=~~2~~ ~~1~~ 0 1 2 val=~~1~~ ~~2~~ ~~3~~ ~~2~~ 1

```
- - 1
- 2 2 2
3 3 3 3 3
  2 2 2
- - 1
```